Computer Practical
Shallow-Water Model

Waves, rotation, instability and large-scale flow

Paul Connolly and Robin Hogan • September 2026

1 Overview

In this practical you will use a two-dimensional shallow-water model to investigate how a simple set of conservation equations can reproduce several important features of geophysical fluid flow. You will begin with gravity waves, then examine the effect of changing fluid depth, barotropic instability and Rossby waves forced by topography. Finally, you will replace the idealised mountain with realistic Earth orography and examine the response of a uniform westerly flow.

The purpose is not to treat the shallow-water equations as a complete model of the atmosphere or ocean. Instead, use each controlled experiment to connect a visible feature of the simulation with a physical mechanism represented by the equations.

Keep a practical record: save each animation using a meaningful filename and write one or two sentences describing the settings and the main result.

By the end of the practical you should be able to:

- relate the propagation speed of a shallow-water gravity wave to $c = \sqrt{gh}$;
- explain why a gravity wave changes as the fluid becomes shallower;
- explain why an unstable balanced flow requires a perturbation before the instability becomes visible;
- describe the roles of rotation, potential-vorticity conservation and the β effect in large-scale waves;
- recognise both the usefulness and the limitations of a shallow-water model as an atmospheric model;
- recognise a numerical-stability failure caused by an excessively large timestep.

2 What the model represents

The model contains a single layer of fluid with depth $h(x, y, t)$ over fixed orography of height $H(x, y)$. The horizontal wind components are u and v. The height of the upper surface is therefore $h + H$ (Figure 1). The model domain is periodic in the zonal (x) direction.

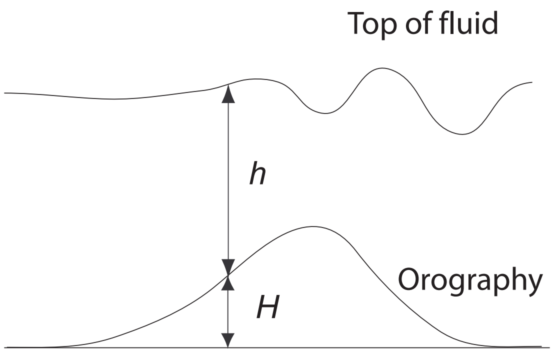

Figure 1: Schematic of the shallow-water model. The prognostic fluid depth is h and the fixed lower-boundary height is H; the plotted upper-surface height is $h + H$.

In the default configuration, the horizontal grid spacing is 100 km and the domain is intended to resemble a mid-latitude channel around the Earth. A circumference at 45° latitude is about $2\pi R_E \cos 45° \simeq 28,000$ km, comparable with the zonal size of the model domain.

The Coriolis parameter varies linearly with y:

$$f(y) = f_0 + \beta(y - \bar{y}), \tag{1}$$

so $f = f_0$ at the centre of the domain. Relative vorticity is

$$\zeta = \frac{\partial v}{\partial x} - \frac{\partial u}{\partial y}. \tag{2}$$

The lower panel of each animation shows ζ; the upper panel shows the upper-surface height together with wind vectors.

For atmospheric experiments, the upper-surface height can be interpreted as a pressure-like or geopotential-like field. This is an analogy: the model still contains only one shallow-water layer and has no explicit vertical atmospheric structure.

3 Model equations and numerical method

The shallow-water model conserves fluid volume and horizontal momentum, so the prognostic variables are h, uh and vh:

$$\frac{\partial h}{\partial t} + \frac{\partial (uh)}{\partial x} + \frac{\partial (vh)}{\partial y} = 0, \tag{3}$$

$$\frac{\partial (uh)}{\partial t} + \frac{\partial (u^2 h + g h^2/2)}{\partial x} + \frac{\partial (uvh)}{\partial y} = h\left(fv - g\frac{\partial H}{\partial x}\right), \tag{4}$$

$$\frac{\partial (vh)}{\partial t} + \frac{\partial (uvh)}{\partial x} + \frac{\partial (v^2 h + g h^2/2)}{\partial y} = h\left(-fu - g\frac{\partial H}{\partial y}\right). \tag{5}$$

The model advances these equations using a two-step Lax–Wendroff scheme. Fluxes are first estimated at intermediate points half a grid spacing and half a timestep from the original grid, and those midpoint fluxes are then used to update h, uh and vh. The Coriolis terms are advanced with a centred Crank–Nicolson update.

Like any explicit numerical scheme, Lax–Wendroff has a stability limit. A useful guide is the Courant number: information should not travel across too many grid boxes in one timestep. The code prints an approximate initial two-dimensional Courant number at the start of every run and warns if it is too large.

4 Log in and download the code

Use the same SSH/SFTP workflow as in *Tools of the Trade*. On campus, the teaching-server address is `130.88.66.57`; otherwise use the server details supplied for the course. Replace `YOUR_USERNAME` with your own username.

Keep two terminal windows open. The **SSH window** is your command line on the teaching server: use it to edit files and run the model. The **SFTP window** is used to copy finished animations from the server back to your own computer.

In the SSH window:

```
ssh -Y YOUR_USERNAME@130.88.66.57
```

In a second, local terminal window:

```
sftp YOUR_USERNAME@130.88.66.57
```

Your password will not be displayed while you type it.

In the SSH window, obtain the model:

```
git clone https://github.com/EnvModelling/shallow-water-model-practical
cd shallow-water-model-practical
```

There is no compilation step: this practical uses the Python version of the model.

File-transfer reminder. In SFTP, `lpwd` shows the folder on *your own computer* into which files will be downloaded. Use `lcd` to change it if required, for example:

```
lpwd
lcd Desktop
```

The general form of the download command is

```
get REMOTE_FILE LOCAL_FILENAME
```

The second filename is the name of the copy on your own computer. Giving each result a different local filename lets you keep several experiments even though the model always uses the same filename on the server.

5 Inspecting and running the model

The settings you change are separated from the numerical code. Open the configuration file with line numbers displayed in the SSH window:

```
nano -l python/config.py
```

The main parameters used below are grouped around lines 26–38. Approximate line numbers are useful with `nano -l`, but they can change when the repository is updated. If a parameter is not where expected, use `Ctrl-W` and search for its name.

Run the model from the repository root in the SSH window:

```
python3 python/shallow_water_model.py
```

The model prints its progress and, when the animation has been created, tells you where it was saved. The output file on the server is always

```
/tmp/YOUR_USERNAME/animation.gif
```

A later run will overwrite this server copy, so download an animation before starting the next experiment. In the SFTP window, give the local copy a useful name at the same time. For example:

```
get /tmp/YOUR_USERNAME/animation.gif gravity.gif
```

This leaves the server file called `animation.gif`, but saves the copy on your own computer as `gravity.gif`. Use the suggested local filename for each later experiment so that you keep all of your results. When the instructions below say “save as `gravity.gif`”, use the same SFTP command pattern with the suggested local filename.

To restore the supplied configuration before a new experiment, use this in the SSH window:

```
git restore python/config.py
```

This discards edits to the configuration file, so save any settings you want to keep in your notes first.

6 Parameters used in the practical

Table 1: Main settings in `python/config.py`. The line numbers are approximate.

Parameter	Approx. line	Supplied value	Meaning
`f`	31	1.0e-4 s^{-1}	Coriolis parameter at the centre of the domain
`beta`	32	1.6e-11 m^{-1} s^{-1}	Meridional gradient of the Coriolis parameter
`dt_mins`	34	1.0 min	Numerical timestep
`output_interval_mins`	35	60 min	Time between animation frames
`forecast_length_days`	36	4 days	Total model integration time
`orography`	38	FLAT	Fixed lower boundary H
`initial_conditions`	39	GAUSSIAN_BLOB	Initial upper-surface-height pattern
`initially_geostrophic`	40	False	Derive winds from geostrophic balance for height-defined initial states
`add_random_height_noise`	41	False	Add a small perturbation to the initial height field
`random_seed`	43	1	Makes random perturbations reproducible between runs

The available named choices include:

- orography: `FLAT`, `SLOPE`, `GAUSSIAN_MOUNTAIN`, `EARTH_OROGRAPHY` and `SEA_MOUNT`;
- core initial states: `GAUSSIAN_BLOB`, `ZONAL_JET` and `UNIFORM_WESTERLY`;
- further initial states: `SHARP_SHEAR`, `SINUSOIDAL` and `EQUATORIAL_EASTERLY`.

The `UNIFORM_WESTERLY` state is constructed from a free-surface slope and then the wind is diagnosed from geostrophic balance. Keep this detail in mind for the Rossby-wave experiment below.

Note. Keep `nx=254` and `ny=50` when using `EARTH_OROGRAPHY`, because the supplied elevation data use that grid.

7 Experiments

For each subsection, restore `python/config.py` before entering the listed settings unless the instructions explicitly say to modify the preceding run. Save the animation before starting the next run.

7.1 Gravity waves and numerical stability

A non-uniform free surface with the winds initially at rest is not in force balance. The resulting pressure-gradient force launches gravity waves. Remove rotation so that the response is as simple as possible. Set:

```
f = 0.0
beta = 0.0
orography = FLAT
initial_conditions = GAUSSIAN_BLOB
initially_geostrophic = False
add_random_height_noise = False
output_interval_mins = 15.0
forecast_length_days = 0.5
```

Run the model and save the animation as `gravity.gif`.

Prediction. For a layer depth of about 10 km, estimate $c = \sqrt{gh}$. Roughly how far should a gravity wave travel in one hour? Use the horizontal axis of the animation to make a prediction before viewing the result.

Exercise. Estimate the propagation speed from the animation and compare it with $\sqrt{gh}$. Is there much signal in the relative-vorticity panel? Why are gravity waves different in this respect from rotational waves?

Optional numerical-stability test. Keep the gravity-wave settings but change only

```
dt_mins = 5.0
```

The output interval of 15 minutes is still an exact multiple of the timestep. Run the model again. The printed Courant number is now greater than one and the integration should become unstable; the updated code stops with a clear error once it produces an invalid fluid depth. This is a diagnostic run rather than a result to save, so there is no useful animation to download. Record the printed Courant number and error message instead.

Exercise. What has changed physically between the 1-minute and 5-minute experiments? What has changed numerically? Why does a numerical-stability failure not represent a real atmospheric or oceanic process?

7.2 A tsunami analogue: changing fluid depth

A tsunami is also a shallow-water gravity wave, but its speed depends on the local ocean depth. Restore the supplied configuration, then use the gravity-wave setup with a submerged Gaussian seamount:

```
f = 0.0
beta = 0.0
orography = SEA_MOUNT
initial_conditions = GAUSSIAN_BLOB
initially_geostrophic = False
add_random_height_noise = False
output_interval_mins = 60.0
forecast_length_days = 4.0
```

Run and save the result as `tsunami.gif`.

Prediction. Since $c = \sqrt{gh}$, what should happen to the wave speed as the wave enters shallower water over the seamount? If the period remains similar, what should happen to its wavelength?

Exercise. Identify the changes in the wave as it encounters the shallower region. Which features resemble tsunami shoaling? Remember that the model uses a highly idealised topography and a much larger surface disturbance than a real tsunami.

7.3 Barotropic instability

Large-scale flows can contain strong horizontal shear. In a shallow-water system, a necessary condition for barotropic instability is that the meridional gradient of potential vorticity,

$$\frac{\partial}{\partial y}\left(\frac{f+\zeta}{h}\right), \tag{6}$$

changes sign somewhere in the flow. Start with an idealised Bickley-like jet:

```
f = 1.0e-4
beta = 1.6e-11
orography = FLAT
initial_conditions = ZONAL_JET
initially_geostrophic = True
add_random_height_noise = True
output_interval_mins = 60.0
forecast_length_days = 4.0
```

Run and save the result as `jet_noise.gif`. The fixed `random_seed` means that all students should apply the same small perturbation.

Now change only

```
add_random_height_noise = False
```

and save the result as `jet_no_noise.gif`.

Prediction. An instability amplifies perturbations, but does it create a perturbation from an exactly symmetric state? Predict which run should develop visible waves first.

Exercise. Compare the two animations. Why does the random perturbation matter even though the underlying jet is unstable? Relate your answer to the distinction between an unstable basic state and the disturbance that grows on it.

Note. If time permits, repeat the noisy experiment with `initial_conditions = SHARP_SHEAR`. Save the result as `sharp_shear.gif`. This produces a more frontal, cyclogenesis-like example of shear instability.

7.4 Orographic Rossby waves

A dominant feature of the Earth's atmosphere is the presence of large-scale Rossby, or planetary, waves. In the Northern Hemisphere these are forced in part by flow over major mountain ranges such as the Himalayas and Rockies. In a rotating shallow-water system, motion over topography changes the fluid depth and therefore the relative vorticity required to conserve potential vorticity. The β effect provides an additional planetary-vorticity restoring mechanism.

Start with an isolated Gaussian mountain. Set

```
f = 1.0e-4
beta = 1.6e-11
orography = GAUSSIAN_MOUNTAIN
initial_conditions = UNIFORM_WESTERLY
initially_geostrophic = True
add_random_height_noise = True
output_interval_mins = 60.0
forecast_length_days = 4.0
```

Run the model and save the animation as `mountain_beta.gif`.

Exercise. Explain the initial meander in terms of vortex stretching and compression as the fluid passes over the mountain. Where is relative vorticity generated, and why?

A deliberately unsuccessful experiment. Now change only

```
f = 0.0
beta = 0.0
```

and run the model again. Do not try to "fix" the result immediately. Inspect what Python prints in the terminal. This is also a diagnostic run rather than a result to save: the initial state is not constructed successfully, so there is no meaningful animation to download.

Exercise. Why do `NaN` values appear? In `python/shallow_water_model.py`, find the section beginning `if cfg.initially_geostrophic`. Look carefully at the expressions used to calculate u and v. Which quantity appears in a denominator, and what value does it have after you set `f = 0.0` and `beta = 0.0`? Explain why the initial state cannot be constructed. Is this the physical response of a non-rotating 20 m s^{-1} westerly, or has the problem occurred before the dynamical experiment can begin?

Note. This is a useful modelling lesson: changing a physical parameter can invalidate an assumption used to construct the initial state. A failed run can therefore tell you something about the model formulation rather than about the subsequent dynamics.

Finally, restore

```
f = 1.0e-4
beta = 0.0
```

with the other mountain settings unchanged. Run the model and save it as `mountain_fplane.gif`.

Exercise. Compare the f-plane run with your original β-plane run. What happens to the disturbance after the air passes over the mountain? Why is the response different when f has no meridional gradient? Think particularly about the restoring mechanism for a Rossby wave.

7.5 Flow over real Earth orography

The Gaussian-mountain experiment isolated the response to a single idealised obstacle. For the final core experiment, keep the same β-plane and uniform westerly flow but replace the Gaussian mountain with the supplied Earth orography. This gives a less controlled, but more recognisable, lower boundary and lets you ask whether the same topographic Rossby-wave mechanism appears in a more realistic setting.

Restore `python/config.py` and set:

```
f = 1.0e-4
beta = 1.6e-11
orography = EARTH_OROGRAPHY
initial_conditions = UNIFORM_WESTERLY
initially_geostrophic = True
add_random_height_noise = False
output_interval_mins = 60.0
forecast_length_days = 4.0
```

Keep `nx = 254` and `ny = 50`. Run the model and save the result as `earth_orography.gif`.

The 1 m orography contour provides an approximate coastline in the animation; the upper panel also shows contours of the higher terrain. Use these to relate the flow response to the position of the continents and major mountain ranges.

Prediction. Before running the model, where would you expect the largest disturbances to the uniform westerly flow: over flat ocean regions, over major topography, or downstream of it? Why?

Exercise. Compare `earth_orography.gif` with your Gaussian-mountain β-plane run. Which features can you relate directly to major topography? Which disturbances remain approximately tied to the orography, and which propagate downstream? Why is the response more complicated than for one isolated mountain?

Exercise. This experiment still uses only one shallow-water layer. Which important aspects of the real atmosphere are therefore absent, even though the horizontal flow can develop recognisable planetary-scale structure?

7.6 Further experiments if time permits

The model contains several additional initial conditions that are useful for exploration after the core comparisons above. Restore the supplied configuration before each one, then change only the settings listed.

Jupiter-like alternating jets

```
initial_conditions = SINUSOIDAL
initially_geostrophic = True
add_random_height_noise = True
```

Run the model and save the result as `jupiter_jets.gif`.

The alternating zonal jets can generate vortices that interact and merge, illustrating the tendency of two-dimensional/geostrophic turbulence to transfer energy towards larger scales. How do the long-lived vortices compare qualitatively with Jupiter's Great Red Spot?

Equatorially trapped waves

```
f = 0.0
beta = 2.5e-11
initial_conditions = EQUATORIAL_EASTERLY
initially_geostrophic = True
add_random_height_noise = True
output_interval_mins = 120.0
forecast_length_days = 8.0
```

Run the model and save the result as `equatorial_waves.gif`.

Because f changes sign across the middle of the domain, disturbances can become trapped near the model equator.

Equatorial Kelvin wave

```
f = 0.0
beta = 5.0e-10
output_interval_mins = 15.0
forecast_length_days = 0.5
```

Run the model and save the result as `kelvin_wave.gif`.

Compare the eastward and westward responses. Which disturbance remains trapped near the equator, and how does rotation make eastward and westward propagation asymmetric?

8 What you should take away

A very small set of conservation laws can reproduce a surprisingly wide range of behaviour. Gravity provides fast restoring waves; changing fluid depth changes their speed. Rotation links horizontal pressure gradients to balanced flow and makes potential-vorticity conservation dynamically important. Sheared balanced flows can amplify perturbations, while topography and the meridional change of planetary vorticity generate large-scale wave responses.

The same practical also illustrates two modelling lessons. First, a model experiment is easiest to interpret when one physical assumption is changed at a time and all other settings are controlled. Second, a failed or strange run is not automatically a physical result: timestep choice and initialization both matter. The $f = 0$ westerly experiment is a particularly clear example because the geostrophic initial condition itself becomes mathematically undefined. A useful model user therefore asks both “what process is represented?” and “could this feature have been produced by the numerical method or the experimental setup?”